

Multiple Choice (10 marks)

1. Among these colors, the one that has the most energy per photon is
 - (a) violet
 - (b) turquoise
 - (c) green
 - (d) blue
 - (e) red
2. The red glow in the neon tube of an advertising sign is a result of
 - (a) fluorescence.
 - (b) absorption.
 - (c) resonance.
 - (d) de-excitation.
 - (e) refraction.
3. A heavy rock and a light rock in free fall (zero air resistance) have the same acceleration. The heavy rock doesn't have a greater acceleration because the
 - (a) air resistance is always zero in free fall.
 - (b) volume of both rocks is the same.
 - (c) force due to gravity is the same on each.
 - (d) gravitational constant is the same for both rocks.
 - (e) force due to gravity is zero in free fall.
4. When small pieces of material are assembled into a larger piece, the combined surface area
 - (a) greatly increases
 - (b) slightly increases
 - (c) is halved
 - (d) becomes zero
 - (e) greatly decreases
5. A corrective lens of refractive power - 10.00 diopters corrects a myopic eye. What would be the effective power of this lens if it were 12 mm. from the eye?
 - (a) -11.00 diopters

- (b) -9.50 diopters
- (c) -10.50 diopters
- (d) -9.00 diopters
- (e) -7.93 diopters

True / False (10 marks)

Answer True or False

6. True or False: An electron confined within a nuclear radius can have a minimum kinetic energy of 8.7 MeV due to its wave nature.
7. True or False: The correct answer to 'Determine the total collisional frequency for CO₂ at 1 atm and 298 K.' is ' $8.44 \times 10^{34} \text{ m}^{-3} \text{ s}^{-1}$ '.
8. True or False: An electron can be effectively accelerated by a magnetic field when it is at rest.
9. True or False: When relatively slow-moving molecules condense from the air, the temperature of the remaining air tends to decrease.
10. True or False: The complementary color of blue in the additive color model is green.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given a projector with a 40 cm focal length illuminating a 4.0 cm object, analyze how the distance from the lens to the screen affects the size of the projected image. Calculate the image size and discuss the underlying optical principles involved.
12. Discuss the process of uranium nucleus fission, focusing on the forms of energy released, particularly the role of alpha particles, and analyze their significance in nuclear reactions.

End of Paper